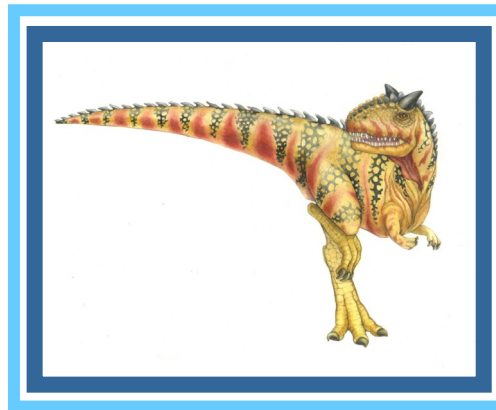
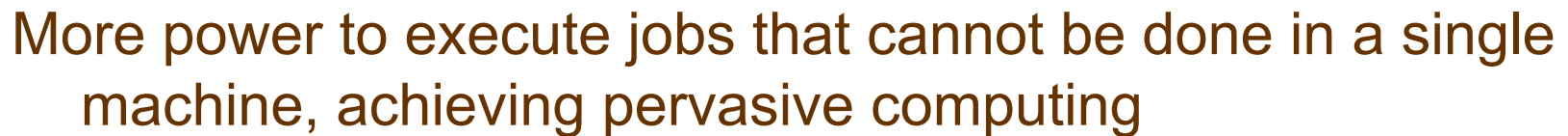


Chapter 16: Distributed System and Cloud Computing





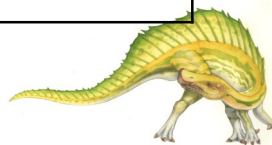
-
- A Venn diagram with three overlapping circles. The top circle is red and labeled 'Computing'. The bottom-left circle is blue and labeled 'Networking'. The bottom-right circle is yellow and labeled 'Storage'. The intersections are shaded: purple for Computing and Networking, orange for Computing and Storage, and green for Networking and Storage. The central intersection of all three is shaded brown.





Selected applications on distributed systems

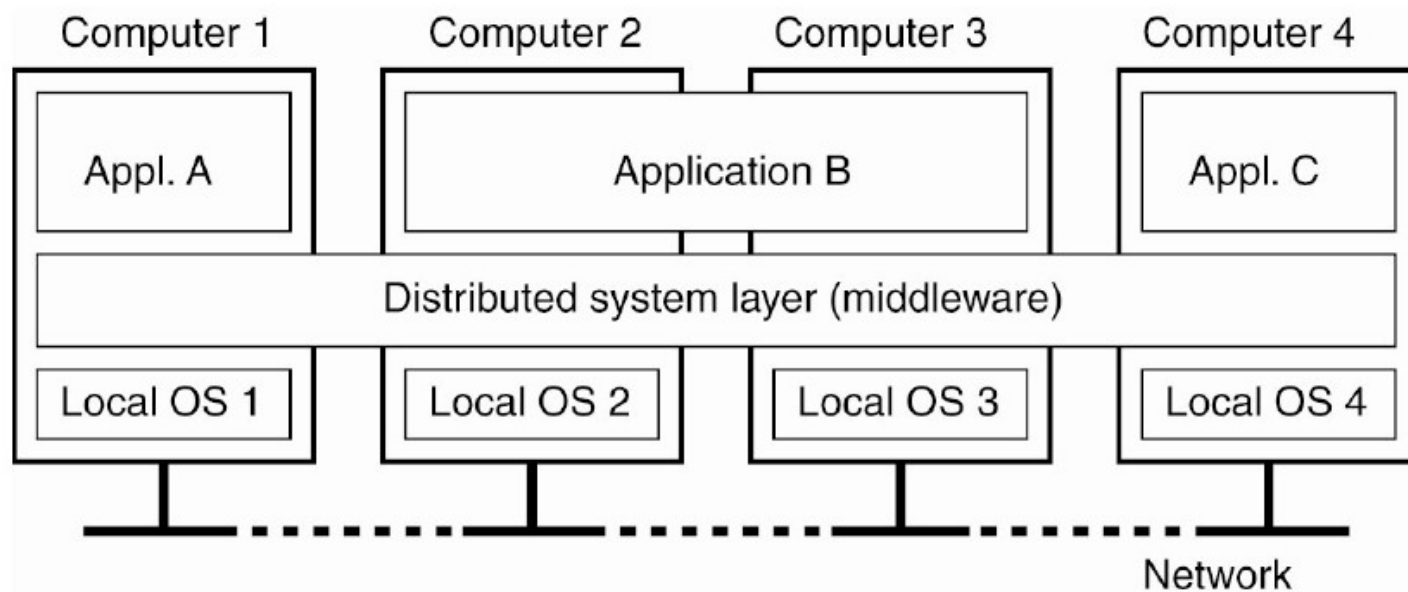
<i>Finance and commerce</i>	eCommerce e.g. Amazon and eBay, PayPal, online banking and trading
<i>The information society</i>	Web information and search engines, ebooks, Wikipedia; social networking: Facebook and MySpace.
<i>Creative industries and entertainment</i>	online gaming, music and film in the home, user-generated content, e.g. YouTube, Flickr
<i>Healthcare</i>	health informatics, on online patient records, monitoring patients
<i>Education</i>	e-learning, virtual learning environments; distance learning
<i>Transport and logistics</i>	GPS in route finding systems, map services: Google Maps, Google Earth
<i>Science</i>	The Grid as an enabling technology for collaboration between scientists
<i>Environmental management</i>	sensor technology to monitor earthquakes, floods or tsunamis





Distributed System (cont.)

- Distributed system is an operating system that manages multiple computer through networks.
- It serves as a middleware that provides interface for multiple applications across multiple machines





Resource sharing

- Resource allocation in a physical infrastructure
- Computing, networking and storage resources
- Efficiency, scalability and reliability/availability

- *Distributed system's perspective*
- Sharing a service, e.g., web search engine
- Read, write and delete operations on files
- An interface that provides and manages these operations
- Remote invocation through communication





Challenges in Distributed System

- Concurrency
 - Concurrent execution of jobs
 - Coordinate actions
- No global clock
 - Synchronization
- Independent failure
 - Hardware/software failure
 - Protection, recovery and availability





What is Cloud Computing?

- **Cloud Computing** is a general term used to describe a new class of network based computing that takes place over the Internet,
 - basically a step on from Grid Computing
 - a collection/group of integrated and networked hardware, software and Internet infrastructure (called a platform).
 - Using the Internet for communication and transport provides hardware, software and networking services to clients
- These platforms hide the complexity and details of the underlying infrastructure from users and applications by providing very simple graphical interface or API (Applications Programming Interface).





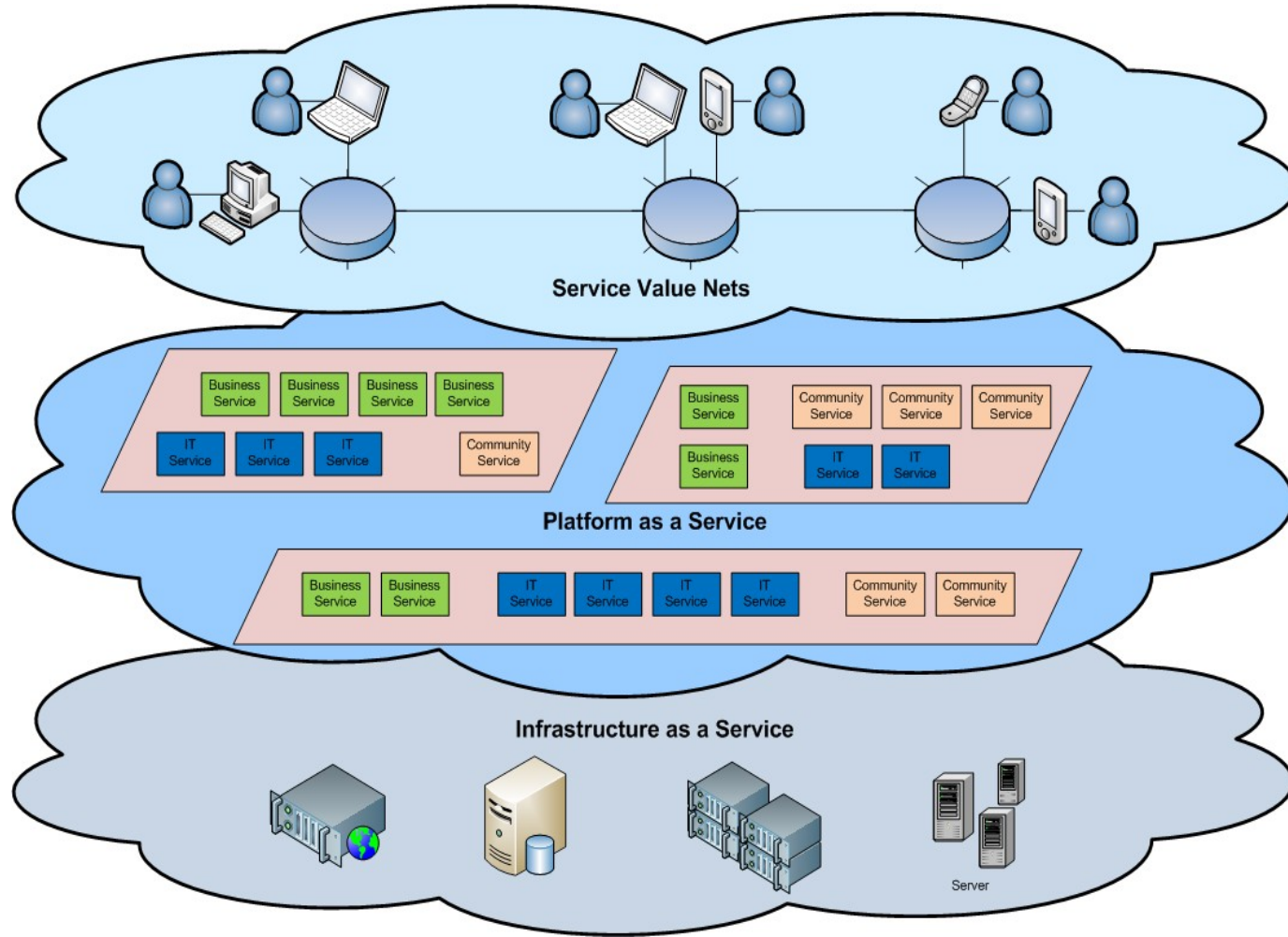
What is Cloud Computing? (cont.)

- In addition, the platform provides on demand services, that are always on, anywhere, anytime and any place.
- Pay for use and as needed, elastic
 - scale up and down in capacity and functionalities
- The hardware and software services are available to
 - general public, enterprises, corporations and businesses markets





Cloud Architecture





Virtualization

□ Virtual workspaces:

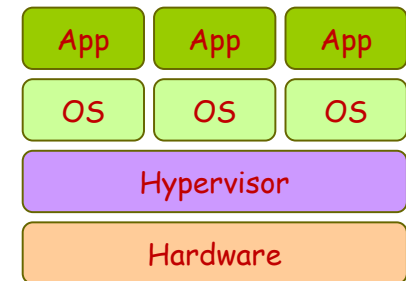
- An abstraction of an execution environment that can be made dynamically available to authorized clients by using well-defined protocols,
- Resource sharing (e.g. CPU, memory share),
- Software configuration (e.g. O/S, provided services).

□ Implement on Virtual Machines (VMs):

- Abstraction of a physical host machine,
- Hypervisor intercepts and emulates instructions from VMs, and allows management of VMs,
- VMWare, Xen, etc.

□ Provide infrastructure API:

- Plug-ins to hardware/support structures

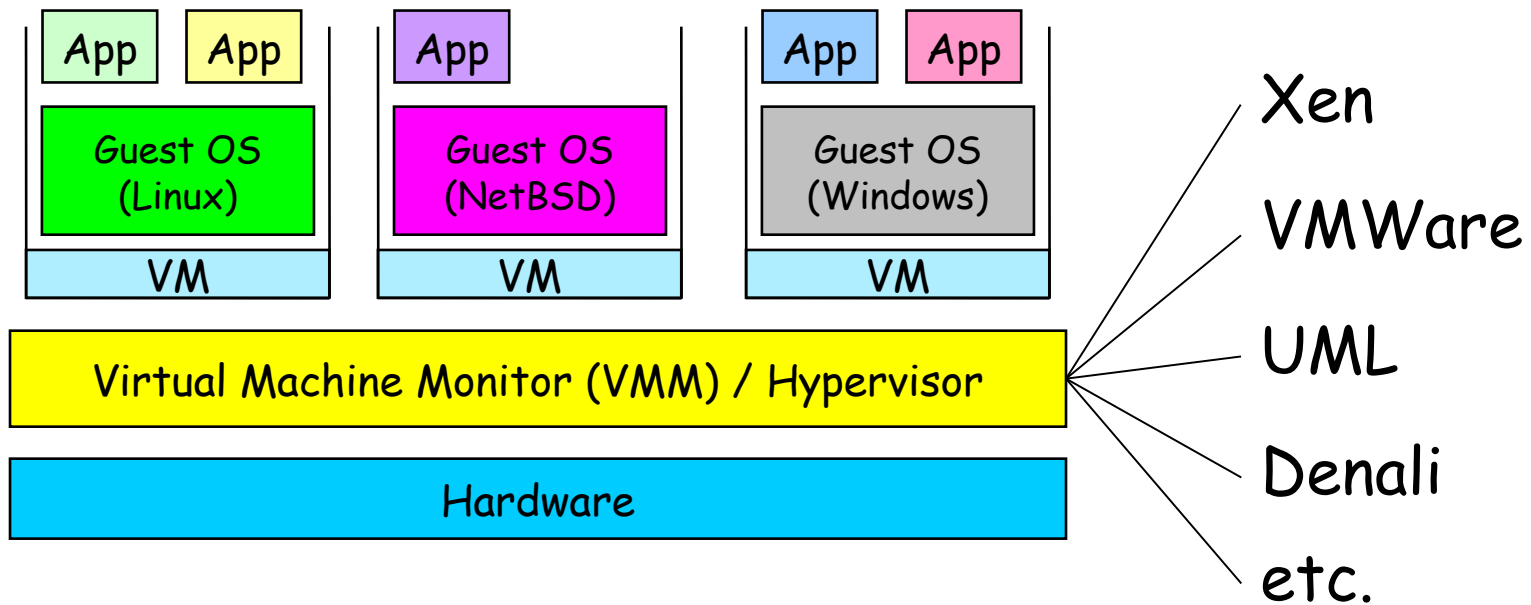


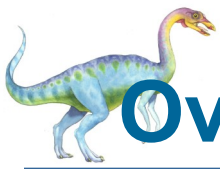
Virtualized Stack





Virtual Machines





Overall Google Systems Architecture

Google applications and services

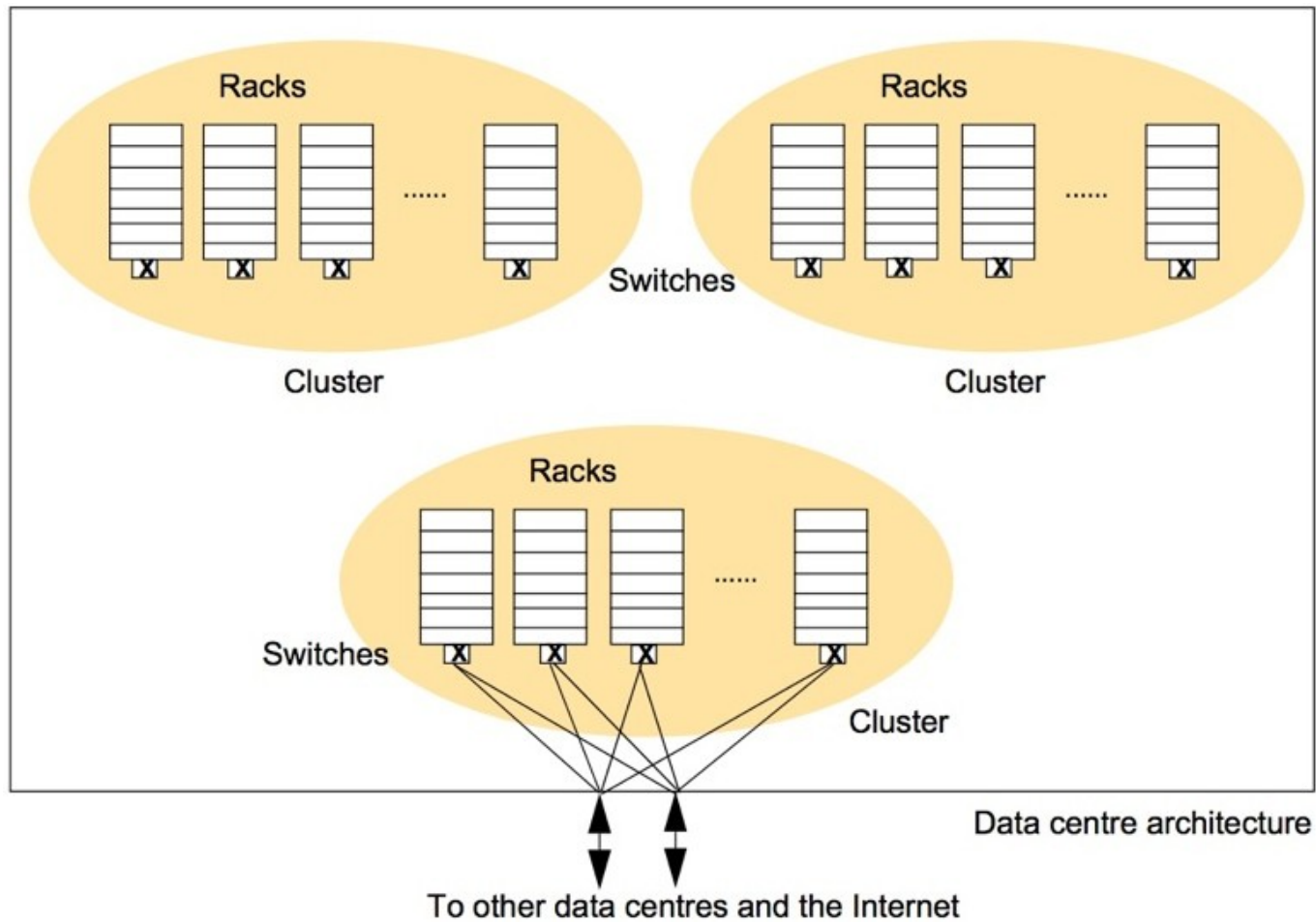
Google infrastructure (middleware)

Google platform



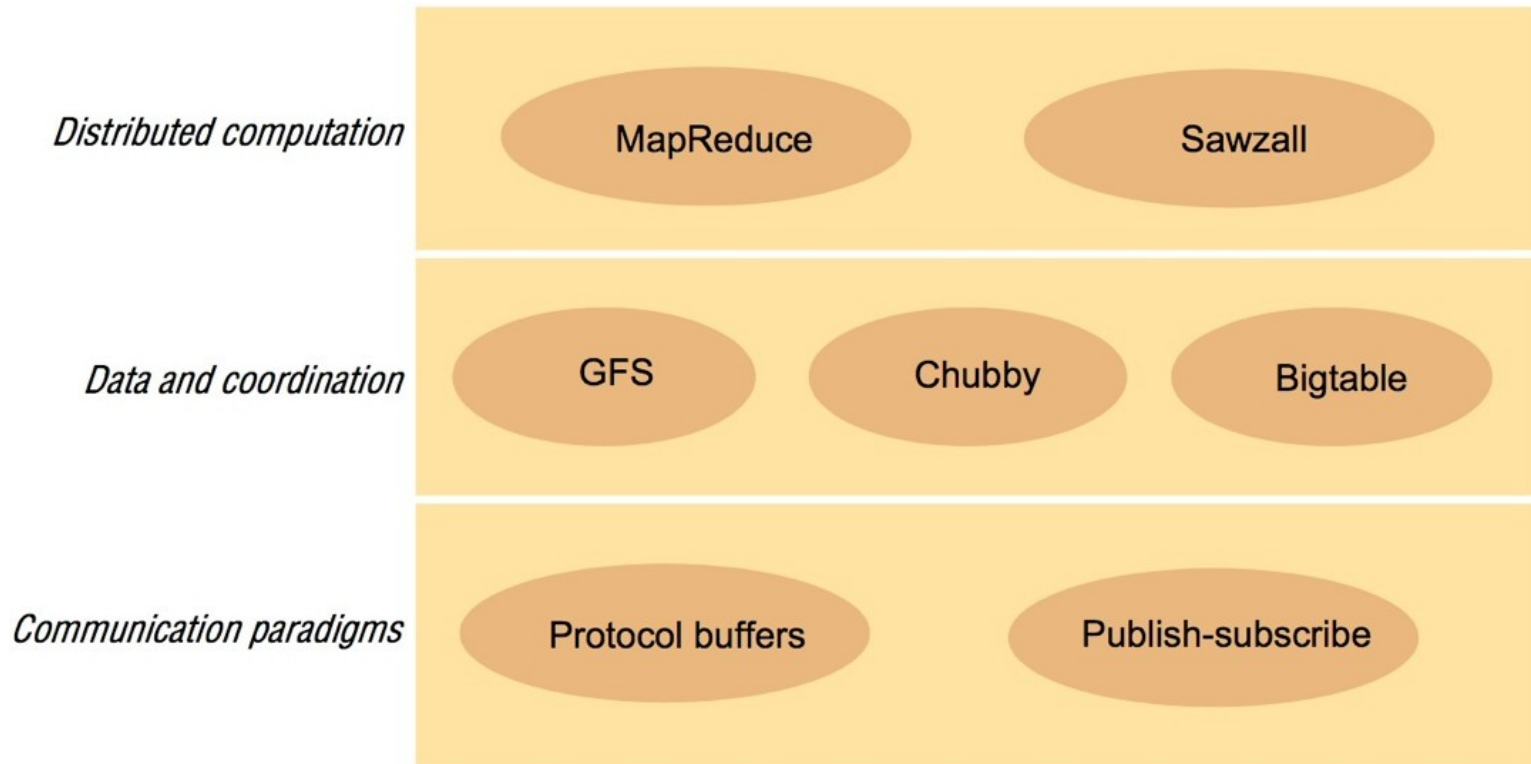


Physical Model of the Google Infrastructure





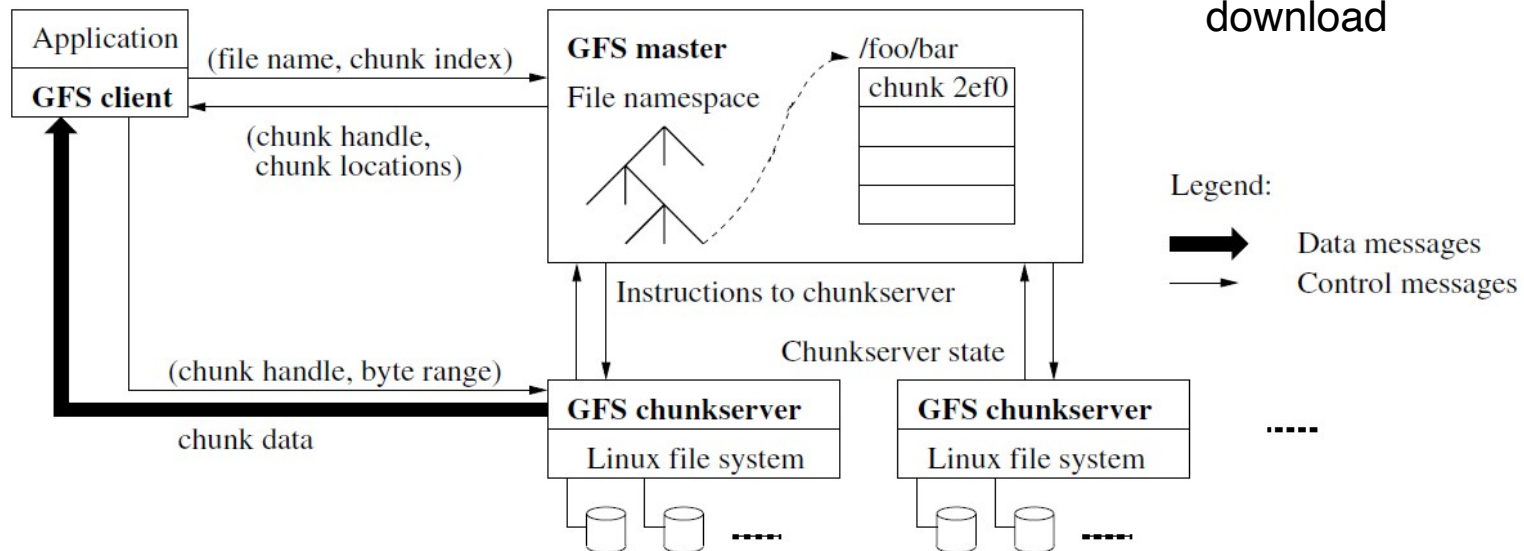
Architecture of Google Infrastructure





Google File System (GFS)

- **Chunk**
 - 64MB fixed size
 - Chunk handle (64 bit unique identification)
 - Default three replicas
- **Single master**
 - File and chunk
 - Mapping from file to chunks
 - Chunk location
- **Chunk location**
 - Masters polls chunkserver for location information
 - Clients contact master for location information and
 - cache it for a while
 - Clients contact chunkserver directly for download





GFS Master

- Metadata
 - File and chunk
 - Mapping from file to chunk
 - Location of each chunk
- Problems
 - Single point of failure, scalability bottleneck
- Solution
 - Shadow master, minimize master involvement
- Operational log
 - Keeps track of activities, providing a logical time line
 - Replicated in multiple sites for reliability





Read and Write in GFS

- Read operation:
 - Simply read from the primary chunkserver
- Write operation:
 - Master picks one replica as primary;
 - Primary defines the mutation order;
 - The other replicas follow the same order;
 - Minimize overhead at the master;
 - Data flow is in a carefully defined linear chain;

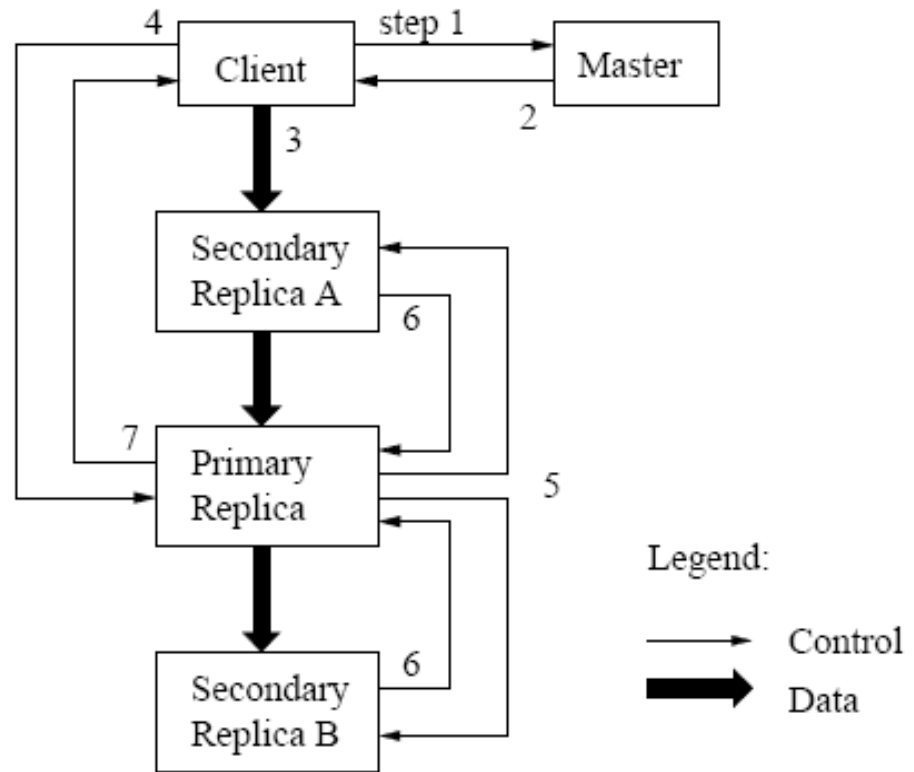


Figure 2: Write Control and Data Flow



End of Chapter 16

